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AI Smartglasses in Practice: Reality and Challenges

Tingting Yao, Ruizhe Shi, Yao Liu, Bo Han, and Songqing Chen

***Abstract*—AI-powered smartglasses are becoming everyday platforms for hands-free sensing, media capture, live streaming, and conversational assistance. Yet little is known about how these commercial systems actually perform in practice, especially because their hardware and implementations are largely closed. This paper presents a measurement-based case study of Ray-Ban Meta smartglasses through three representative functions: AI Interaction, Quick Capture, and Live Streaming, together with the Quality of Experience delivered by its voice AI assistant.**

Our measurements reveal three practical design challenges. First, the glasses keep Bluetooth available for conversational AI but activate Wi-Fi Direct only when transferring photos and videos. Although this division accommodates different data rates, establishing Wi-Fi Direct adds a mean latency of 3.49 s before a Quick Capture transfer can begin, and the cost is repeated when successive transfers cannot reuse the connection. Second, Internet access is mediated by the paired phone. During Live Streaming, the glasses-phone and phone-server connections use TCP and QUIC, respectively, and maintain their connection states independently. For example, in one representative trace, a user-visible stall coincided with a 3.03 s interruption of the server-facing connection even though the glasses-facing connection remained active. This shows that the phone must coordinate the two hops rather than simply relay traffic. Third, across 100 voice queries, Ray-Ban Meta produced much shorter responses than Meta AI on the phone but exhibited 70 percent higher user-perceived latency and substantially lower quality on open-ended tasks. These findings motivate connection reuse, coordinated two-hop recovery, and task-adaptive AI responses for future smartglasses.

INTRODUCTION

AI-powered smartglasses are moving from experimental prototypes to consumer devices designed for everyday use. Products from Meta, Rokid, RayNeo, etc. integrate cameras, microphones, speakers, wireless connectivity, and cloud-based AI into a conventional eyewear form factor. Recent advances in large language models have further expanded their capabilities from media capture and voice commands to conversational assistance, visual understanding, and hands-free task support. As a result, smartglasses are becoming an emerging platform for continuous sensing, multimodal interaction, and ubiquitous access to AI services.

Smartglasses combine first-person multimodal sensing and interactive AI within a single wearable platform. Compared with smartwatches, which typically exchange intermittent notifications and sensor data, and Bluetooth earbuds, which primarily serve as audio peripherals to a phone, smartglasses generate more diverse traffic through their cameras, microphones, and AI services. They must simultaneously support latency-sensitive voice interactions, bursty transfers of captured photos and videos, and sustained live video streams. This workload diversity imposes substantially different requirements for immediate availability, transfer completion time, streaming continuity, and resource efficiency.

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Like many consumer wearables, smartglasses are known to operate with a paired phone. However, the existence of this pairing does not fully reveal how the glasses access Internet services. It remains unclear whether the glasses can establish a direct Internet path for any function or whether all cloud-bound traffic must pass through the paired phone. Further questions arise regarding its networking role: which local wireless interfaces carry different types of traffic, whether the phone simply provides connectivity or terminates and re-establishes transport connections, what protocols are adopted, and how the glasses-facing and server-facing paths are coordinated. These architecture and protocol choices can directly shape service responsiveness, continuity, resource efficiency, and ultimately the user-perceived Quality of Experience (QoE). Nevertheless, despite the rapid commercialization of AI smartglasses, little is publicly known about their underlying networking architecture and performance.

To address this gap, we present controlled, systematic measurements and performance characterization of commercial AI smartglasses, using the Ray-Ban Meta Gen 1 (hereafter referred to as RayBan) [1] as a case study. We select RayBan because it supports AI Interaction, Quick Capture, and first-person Live Streaming within a single deployed platform, allowing us to study interactive, on-demand, and continuous traffic under the same device ecosystem. Notably, the Meta AI app (hereafter referred to as Meta AI) [2] serves two roles: it is the companion app used to pair and manage the glasses, and it provides direct access to Meta AI on the phone. We therefore use phone-side Meta AI as the smartphone baseline for our QoE evaluation. This ecosystem-matched comparison avoids introducing an unrelated AI provider while allowing us to examine how glasses-specific models and response strategies affect the user experience.

Specifically, we first characterize phone-mediated Internet access and local interface selection by tracing AI Interaction, Quick Capture, Live Streaming, and quantifying the startup latency and glasses-phone transfer rate of the observed Bluetooth and Wi-Fi Direct paths. Second, we examine two-hop protocol coordination during Live Streaming by temporally aligning the TCP traffic on the glasses-to-phone (G2P) hop with the QUIC traffic on the phone-to-server (P2S) hop, focusing on media-source switching and connection recovery. Third, we evaluate AI QoE by comparing RayBan with phone-side alternative Meta AI across 100 factual and open-ended voice queries in terms of response quality, response length, and user-perceived latency. Because the application-layer traffic is encrypted, our network analysis relies on observable packet timing, sizes, directions, endpoints, interfaces, and Bluetooth HCI logs rather than payload inspection. Our measurements

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reveal the following findings.

- **Internet access is phone-mediated, while local interface selection is function-specific.** For all three functions examined in our study, the glasses do not establish a direct path to Internet services; instead, all cloud-bound traffic is relayed through the paired phone. The local glasses-phone path, however, differs across functions. RayBan keeps Bluetooth Low Energy (BLE) available for AI Interaction, whereas Wi-Fi Direct is established on demand for Quick Capture and Live Streaming. For Quick Capture, Wi-Fi Direct provides a high data transfer rate after establishment, but its setup introduces an average delay of 3.49 s before media transfer can begin. This overhead is repeated when a subsequent import cannot reuse the existing connection. The current design is therefore function-specific, but it does not fully exploit connection state across successive operations.
- **The paired phone forms a two-hop gateway, but the two transport hops maintain their states independently.** Live Streaming uses TCP over Wi-Fi Direct on the glasses-phone hop and QUIC on the phone-server hop. Our repeated experiments show that the phone can retain the existing P2S session during a planned switch from phone-originated to glasses-originated video. However, a representative trace from the ten experiments also captures a recovery event in which the P2S connection was interrupted for 3.03 s while the G2P connection remained active and continued delivering data to the phone. This cross-hop state divergence shows that the phone cannot be treated as a transparent relay. It must instead coordinate backpressure, source-rate adaptation, and connection recovery across two independently maintained transport hops.
- **A fixed brevity-oriented AI strategy does not provide consistent QoE across task types.** RayBan produces substantially shorter responses than Meta AI, consistent with a glasses-oriented brevity optimization. However, its mean user-perceived latency remains 70 percent higher, showing that this optimization is insufficient to close the responsiveness gap. Although the strategy is effective for Factual QA, where RayBan achieves quality close to Meta AI (88.0 percent versus 92.0 percent), it becomes counterproductive for Open-Ended QA, where reduced explanations and supporting details coincide with a 44.0-percentage-point quality gap.

Taken together, these findings show that current AI smartglasses adopt function-specific optimizations but lack state and task awareness. Their networking stack neither fully reuses connection state nor coordinates two-hop recovery, while their AI applies a fixed brevity strategy across tasks. Future paired phones should act as adaptive coordinators that anticipate workloads, reuse local links, coordinate cross-hop recovery, and enable task-adaptive AI responses that balance latency and content depth.

Background

Smartglasses are wearable devices that integrate one or more advanced features compared to corrective glasses or fashion glasses, such as AI assistants, Bluetooth connectivity, network access, embedded cameras, speakers, microphones, and displays. These functionalities enable smartglasses to support a wide range of applications, including real-time communication [3], multimedia capture [4], contextual information retrieval [5], and even AR experiences [6]. Unlike cumbersome headsets, smartglasses offer a more portable and socially acceptable form factor, making them particularly attractive for everyday use.

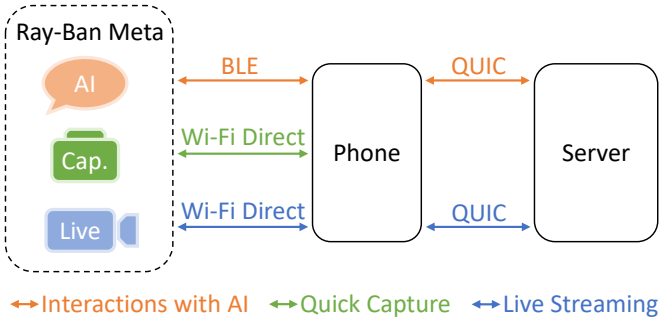


FIGURE 1. Three key functions of RayBan using different networking interfaces and protocols.

Recent advances in sensors, low-power computing, and wireless communication technologies have further accelerated the innovation and adoption of smartglasses in the consumer market. Among these, RayBan glasses distinguish themselves as a leading example of AI-powered smartglasses. Developed by Meta, they are commercially available and widely adopted. Although not equipped with displays, the glasses have a camera, microphones, and speakers, integrating several key functions such as quick photo/video capture, live streaming to social media platforms, hands-free voice commands, and interaction with AI assistants.

Related Work Existing works on smart wearables have spanned several dimensions, including user authentication [7], interaction design [8], energy characterization [9], understanding their social impact by user study [10] , and phone-mediated networking [11], [12]. However, none of them have studied the performance and system design of smartglasses.

Network Flow, Interfaces, and Protocols

As shown in Fig. 1, RayBan offers three key functions: (1) interacting with cloud-based AI assistants via voice commands, without or with accompanying pictures, which we refer to as **AI Interaction**; (2) capturing photos/videos and transmitting (or importing) them from the glasses to the paired smartphone, which we refer to as **Quick Capture**; (3) live streaming from the glasses' camera to a Meta social media app such as Instagram, which we refer to as **Live Streaming**. In this section, we characterize the network flows, interfaces, and transport protocols supporting these functions.

We use rooted Google Pixel 8a smartphones as companion devices. We access the phones and execute commands through Android Debug Bridge (ADB). IP traffic is captured using tcpdump, while Bluetooth traffic is recorded using Android's HCI snoop logging. We then parse and analyze both sets of traces in Wireshark. Because the application traffic remains

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TABLE 1. Startup latency and workload-specific trace-derived G2P data rates ($n = 30$).

Interface / Function	Startup Latency (s)			Data Rate (Mbps)		
	Mean	SD	95% CI	Mean	SD	95% CI
BLE / AI Interaction	Pre-established	–	–	0.025	0.006	[0.023, 0.027]
Wi-Fi Direct / Quick Capture	3.49	0.33	[3.37, 3.61]	116.5	16.6	[110.3, 122.7]

encrypted, we reverse-engineer the communication workflow using observable metadata, including packet timing, sizes, directions, and endpoints.

Network Flow

Quick Capture. We observe that when Wi-Fi is disabled and only Bluetooth remains active, Quick Capture becomes non-functional, which suggests that a Wi-Fi-based connection is required for media transfer after capturing. To investigate this behavior, we capture all network packets transmitted to and from the phone when the import operation is in progress and when it is not in progress. We find that during import, the phone initiates a temporary network interface, denoted as *p2p-wlan0-x* (where x represents an integer starting from 0), which indicates the establishment of a Wi-Fi Direct connection. In this connection, TCP is used. IP addresses are assigned within a peer-to-peer (P2P) group, where the phone consistently uses the address 192.168.49.1 as the group owner, and the glasses are assigned addresses in the range of 192.168.49.x.

AI Interaction. As smartglasses, RayBan is equipped with AI support, enabling users to interact with a cloud-based AI assistant via natural conversations, with or without accompanying images. Users can ask two kinds of questions: (1) voice-only questions; and (2) voice questions with pictures captured by the glasses' camera, which are transmitted to the server and then analyzed as well.

We also identify the connection between the phone and the server, where the QUIC protocol is used. However, the Wi-Fi interface on the phone does not present any data transmission between the glasses and the phone. By analyzing the Bluetooth traces, we confirm that audio transfer between the glasses and the phone is over Bluetooth Low Energy (BLE).

Live Streaming. RayBan also supports hands-free, first-person Live Streaming, in which video captured by the glasses' camera is relayed through the paired phone and broadcast via Meta platforms such as Instagram. We find that the connection established between the phone and the Live Streaming server is based on the QUIC protocol, similar to that in AI Interaction, whereas the connection between the glasses and the phone is established using TCP over Wi-Fi Direct, consistent with the behavior observed during Quick Capture.

Architectural Implications. For the three functions examined, we observed no direct Internet path from RayBan; instead, all cloud-bound traffic is relayed through the paired phone. Compared with direct service access from a smartphone, this phone-mediated architecture introduces an additional glasses-phone hop and requires the system to select a local wireless interface for that hop. Across the three functions, this architecture imposes two core requirements: (1) low end-to-end latency despite the additional glasses-phone hop and (2) efficient coordination across heterogeneous interfaces and protocols.

Phone-Mediated Internet Access and Interface Selection

As shown in Fig. 1, Bluetooth is employed for AI Interaction, whereas Wi-Fi Direct is used for Quick Capture and Live Streaming. These function-specific choices motivate us to investigate the underlying tradeoffs between the two interfaces.

To characterize the deployed interface choices, we measure startup latency and the trace-derived G2P data rate under their corresponding workloads. BLE is measured during AI interaction, whereas Wi-Fi Direct is measured during Quick Capture. Because these workloads differ in traffic patterns and transfer sizes, the reported rates characterize the respective deployments. For Quick Capture, startup latency is defined as the interval from initiating media import to completing the TCP three-way handshake over Wi-Fi Direct. For Wi-Fi Direct, the active transfer interval extends from the first to the last G2P TCP packet carrying data for the import. The rate is calculated as $8B/T$, where B is the sum of captured G2P TCP packet lengths within this interval and T is its duration. We use the Wireshark field frame.len, which includes the Linux cooked-capture header and the captured IP/TCP headers. Captured TCP retransmissions are retained rather than deduplicated. Connection establishment is excluded from the rate calculation and reported separately as startup latency. For BLE, we identify the glasses–phone connection in the HCI snoop logs using Wireshark and calculate the rate from the total logged packet bytes in the glasses-to-phone direction over the measurement interval, including the protocol headers present in the logged packets. Because the BLE link remained connected throughout the evaluated AI interactions, connection establishment did not occur within the measurement window. We therefore report the startup latency as pre-established rather than assigning it a value of zero.

Tab. 1 reveals that the two interfaces provide different benefits for their corresponding applications. AI Interaction follows an interactive conversational request–response pattern and carries relatively little audio and control traffic. Its pre-established BLE connection therefore prioritizes immediate availability over transfer rate, with an observed G2P data rate of 0.025 Mbps (SD: 0.006 Mbps; 95 percent CI: 0.023–0.027 Mbps; $n = 30$). Quick Capture, in contrast, transfers media to the paired phone for local access, making total completion time more important than connection startup alone. Wi-Fi Direct provides a substantially higher observed G2P data rate of 116.5 Mbps (SD: 16.6 Mbps; 95 percent CI: 110.3–122.7 Mbps; $n = 30$), but only after an average startup latency of 3.49 s (SD: 0.33 s; 95 percent CI: 3.37–3.61 s; $n = 30$). This fixed delay is incurred before each on-demand

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import begins transmitting data and cannot be amortized when successive imports do not reuse an existing connection. Thus, although the interface assignment matches the broad traffic characteristics of the two functions, the current deployment is function-specific but not state-aware.

Design Implication. Keeping Wi-Fi Direct continuously active would avoid repeated setup but introduce a different limitation. RayBan is officially rated for up to four hours of moderate use [13]. However, in our exploratory tests, continuous execution of the Wi-Fi Direct-based workload could exhaust the glasses battery within approximately one hour, suggesting that continuously maintaining this workload may impose a substantial battery cost. This motivates state- and workload-adaptive interface orchestration: the persistent BLE connection could serve as a control channel to prepare Wi-Fi Direct before an anticipated import, while an established Wi-Fi Direct connection could be retained for a short reuse window to support adjacent transfers and then released according to inactivity and battery state. Such coordination would amortize startup latency without incurring the cost of continuously maintaining Wi-Fi Direct. Because the closed hardware does not allow radio consumption to be isolated from the camera, encoder, and other active components, controlled interface-level energy characterization remains future work.

Two-Hop Protocol Coordination through the Phone Gateway

As illustrated in Fig. 1, Live Streaming follows a two-hop path through the paired phone. The glasses-phone hop uses TCP over Wi-Fi Direct, while the phone-server hop adopts QUIC. This heterogeneous protocol stack motivates us to examine how the phone coordinates the two hops and how the gateway-mediated path affects end-to-end performance.

Measurement Setup. We repeat the Live Streaming experiment ten times under the same configuration at different times to verify the reproducibility of the observed two-hop behavior. In each run, we simultaneously capture the Wi-Fi Direct and Internet-facing traffic at the paired phone, placing packets from both hops on a common timeline. The G2P flow is identified by the TCP connection between the glasses and the phone's P2P interface, while the P2S flow is identified by the QUIC connection from the phone to the streaming server. Captured packet sizes are aggregated over one-second windows to obtain the trace-derived data rate of each hop. Our objective here is to analyze the temporal ordering of discrete protocol events, including source switching, TCP establishment, and QUIC recovery, rather than to estimate an average traffic curve. Because these events occur at slightly different times across runs, averaging or overlaying all ten traces would obscure their ordering and short-lived transitions. The source-switching sequence was observed in all ten runs, whereas a P2S interruption or reconnection event was observed in six of these runs. Figure 2 presents one run illustrating both source switching and a subsequent P2S reconnection. The 3.03-s packet gap reported for this trace is an illustrative observation, rather than an estimate of typical recovery duration, which varies at second-level.

Source Switching across the Two Hops. In the representative run shown in Fig. 2, the stream was initiated

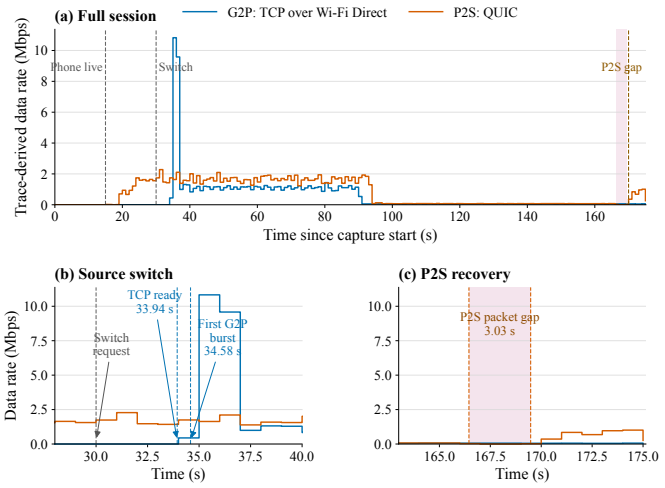


FIGURE 2. Two-hop traffic coordination in a representative Live Streaming run. A single trace is shown to preserve the temporal ordering of short-lived protocol events that would be obscured by averaging multiple runs. The source-switching sequence was observed in all ten runs, whereas P2S interruption or reconnection events were observed in six of the ten runs. The 3.03 s gap refers specifically to the displayed run. The near-baseline curves between 95 and 166 s indicate continuing low-rate traffic on both hops.

from the phone at approximately 15 s, and the source was switched to the glasses at approximately 30 s. Fig. 2(a) shows the complete streaming session, while Fig. 2(b) enlarges the source-switching interval. After the switch was requested at approximately 30 s, the first G2P TCP handshake completed at 33.94 s, followed by the first high-volume G2P packet burst at 34.58 s. During this transition, the existing P2S QUIC connection remained active: between 28 and 40 s, its maximum packet gap was only 176 ms, and its transport four-tuple did not change. The phone therefore preserved the server-facing QUIC session while attaching the newly established G2P TCP path. This behavior demonstrates that source switching is coordinated at the phone gateway rather than implemented by restarting the complete end-to-end streaming session. Nevertheless, the glasses source did not begin delivering substantial G2P traffic until approximately 4.58 s after the switch request, leaving considerable room to reduce source-switching delay.

Cross-Hop Recovery. Although both curves in Fig. 2 appear close to the baseline after approximately 95 s, packet-level inspection confirms that traffic continued on both hops. Between 95 and 166 s, the mean trace-derived G2P and P2S rates were 0.057 and 0.074 Mbps, respectively. This interval is distinct from the subsequent 3.03 s P2S packet gap between 166.45 and 169.48 s. The two transport sessions exhibited different behavior during the observed stall. As shown in Fig. 2(c), the original P2S QUIC flow stopped at 166.45 s, and a new QUIC flow to the same server began at 169.48 s with a different source port, producing a 3.03 s P2S packet gap. In contrast, the G2P TCP connection remained active throughout this interval, carrying 27 packets and 21.8 KB of captured traffic. The observed packet sequence suggests a session-level recovery process. The original P2S connection

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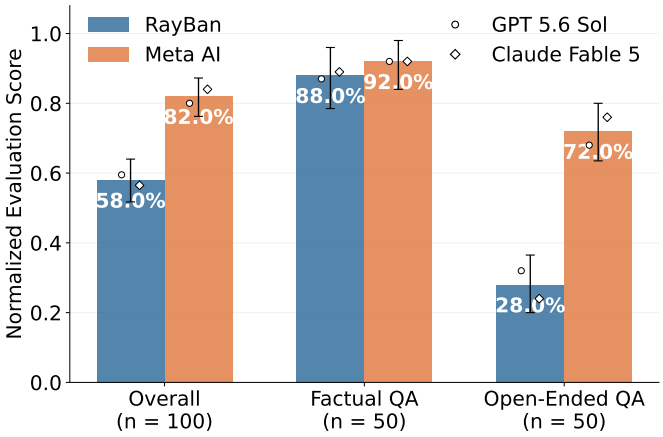


FIGURE 3. Response quality comparison between RayBan and Meta AI. Bars show the equal-weight scores from GPT 5.6 Sol and Claude Fable 5, with error bars indicating question-level bootstrap 95 percent confidence intervals. Circles and diamonds denote the individual judges' scores. RayBan performed comparably to Meta AI on Factual QA but showed a substantial gap on Open-Ended QA, resulting in a lower overall score.

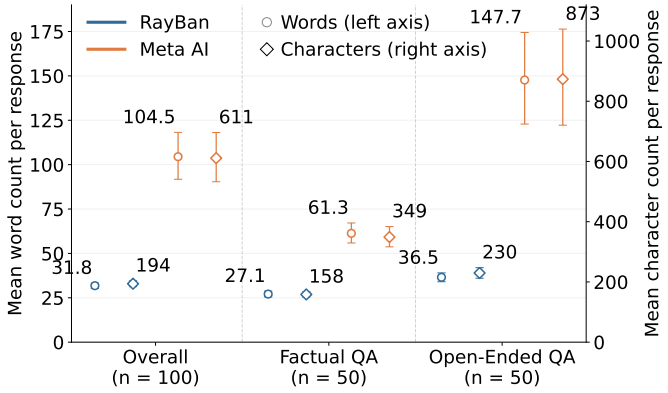


FIGURE 4. Response length comparison between RayBan and Meta AI. Circles show mean word counts (left axis), and diamonds show mean character counts (right axis); error bars indicate question-level bootstrap 95 percent confidence intervals.

exchanged packets in both directions immediately before the interruption, other Internet-facing flows remained active during the gap, and the server responded to the new QUIC Initial within approximately 14 ms. A plausible explanation is that the phone's streaming client recreated the connection following an application-level error or server-initiated session termination. The 3.03 s gap may therefore include session recovery or retry delays, although the encrypted trace does not reveal the exact trigger. This interpretation motivates coordinating P2S recovery with G2P buffering and source-rate control, allowing the phone to manage incoming traffic while the streaming session is re-established.

Design Implications. The G2P TCP connection and P2S QUIC connection are maintained as two independent transport sessions, each with its own connection state and recovery procedure. The current gateway preserves the P2S session during a planned source switch, but this independence also allows the two hops to enter inconsistent states: during the observed P2S disruption, G2P traffic continued arriving at the phone while the server-facing QUIC path was being re-established. Future phone gateways should therefore coordinate transport state in addition to relaying payloads. Before a planned source switch, the phone could pre-establish the G2P path while retaining the existing QUIC session. During a P2S interruption, it could apply bounded buffering, propagate backpressure to the glasses, temporarily adjust the source rate, and prioritize QUIC recovery. Such cross-hop coordination could reduce switching delay, prevent unbounded data accumulation at the phone, and improve streaming continuity without requiring the glasses to access the server directly.

QoE Analysis of AI Interaction

Next, we examine the user-facing QoE of AI Interaction. Specifically, we evaluate response quality, response length, and user-perceived latency, which jointly determine whether an AI assistant can provide timely and useful support through smartglasses. We select the Meta AI on the paired phone as the comparison baseline because both assistants operate within the same Meta AI application ecosystem, enabling an in-ecosystem comparison between the glasses and smartphone interfaces. This comparison does not assume that they use identical underlying models or response policies; indeed, the following experiments reveal distinct response behaviors and optimization strategies. Our objective is therefore to quantify how the two device-specific implementations differ in the end-to-end experience delivered to users.

Question-set construction. We consider two representative voice-assistant scenarios: (1) short, goal-directed factual information seeking, such as asking for a location, date, named entity, or definition; and (2) open-ended conversational practical advice, explanation, planning, and creative generation. To represent these two scenarios, we constructed a dataset of 100 queries, consisting of two 50-query evaluation subsets from VoiceBench [14]: *Factual QA* and *Open-Ended QA*. The Factual QA set contains 50 English questions randomly selected from the SD-QA inventory, retaining the human-written reference answer for each question. Because these questions have well-defined and verifiable answers, responses are evaluated primarily for factual correctness. The Open-Ended QA set contains 25 CommonEval and 25 WildVoice questions, which have no unique reference answer. Although the question texts were drawn from human-recorded VoiceBench subsets, all prompts were re-synthesized using an identical text-to-speech configuration. During testing, audio was played in a quiet environment from the same device at a fixed volume and distance from each system under test, thereby controlling for speaker- and acoustic-input variability.

Response-quality evaluation. Each of the 100 questions was presented to both AI assistants from RayBan and Meta AI, yielding 200 responses. GPT 5.6 Sol and Claude Fable 5 independently evaluated these responses using task-specific criteria. For Factual QA, each response was assessed against

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the reference answer and, when necessary, verifiable authoritative sources, receiving 1 point if correct, 0.5 points if partially correct, and 0 points if incorrect. For Open-Ended QA, the two system responses to each question were compared in terms of relevance, factual soundness, instruction following, and completeness. The preferred response received 1 point, each response received 0.5 points in a tie, and the less-preferred response received 0 points. The two judges were weighted equally by averaging their scores within each question and system.

We report Factual QA and Open-Ended QA separately to characterize task-dependent performance. We also report an Overall score across all 100 questions as a summary of response quality. Because it combines factual-correctness ratings with open-ended pairwise-preference ratings, the Overall score represents a composite normalized evaluation score rather than an accuracy measure. As the two subsets contain equal numbers of questions, it is equivalent to the equal-weight mean of their respective scores.

All statistical analyses treated the question as the independent unit. We used 200,000 question-level nonparametric bootstrap replicates to estimate 95 percent confidence intervals for each system score and the paired difference between RayBan and Meta AI. Directional differences between the systems were evaluated using an exact sign test after excluding tied questions. All randomized analyses used a fixed seed. **Response Quality.** Fig. 3 reveals a clear task-dependent difference in response quality between RayBan and Meta AI. Across all 100 questions, RayBan achieved an equal-weight composite score of 58.0 percent, compared with 82.0 percent for Meta AI, yielding an overall difference of 24.0 percentage points (95 percent CI: 14.5–33.0 percentage points). The subset-level results further show that this aggregate difference was primarily associated with Open-Ended QA rather than Factual QA. For Factual QA, RayBan and Meta AI achieved composite scores of 88.0 percent and 92.0 percent, respectively. The observed difference was 4.0 percentage points (95 percent CI: -3.0 to 12.0 percentage points; exact sign-test $p = 0.6875$). At the question level, the judges reported ties for 44 of the 50 questions, while Meta AI and RayBan were preferred on four and two questions, respectively. These results show that RayBan remained close to Meta AI on the evaluated factual tasks.

The performance pattern changed substantially for Open-Ended QA. RayBan achieved a composite score of 28.0 percent, compared with 72.0 percent for Meta AI, producing a difference of 44.0 percentage points (95 percent CI: 27.0–60.0 percentage points; exact sign-test $p = 1.291 \times 10^{-5}$). The performance gap between Open-Ended QA and Factual QA reveals a pronounced task-dependent performance pattern. Specifically, RayBan approached Meta AI's performance on factual information-seeking tasks but exhibited a pronounced disadvantage on tasks requiring explanation, advice, planning, or other open-ended responses.

Response length provides additional context for the quality results. As shown in Fig. 4, RayBan generated substantially shorter responses than Meta AI overall (31.8 versus 104.5 words). This pattern appeared in both Factual QA (27.1 versus 61.3 words) and Open-Ended QA (36.5 versus 147.7 words), with character counts showing the same trend. The consistently shorter responses reflect a wearable-oriented optimization that prioritizes concise spoken interaction.

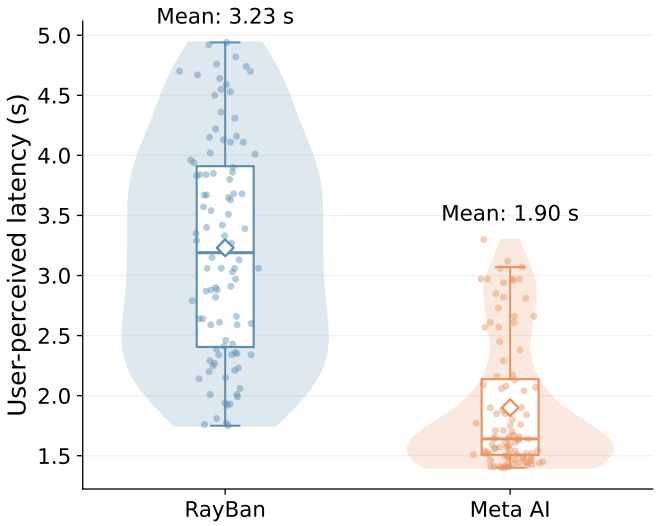


FIGURE 5. User-perceived latency for AI Interaction of RayBan and Meta AI. Violin shapes show the latency distributions across 100 measurements per system; dots indicate individual measurements, boxes show the interquartile ranges with median lines, and diamonds denote the means.

User-perceived latency. We define user-perceived latency as the interval from the end of the user's question to the onset of the assistant's first audible response. As shown in Fig. 5, each system was measured 100 times. RayBan exhibited approximately 70 percent higher mean latency than Meta AI (3.23 s vs. 1.90 s). Its latency distribution was also shifted upward and more dispersed, showing that the higher mean was not caused solely by a few isolated measurements.

Design Implications. These results show that shorter responses alone are not enough to improve the responsiveness of RayBan, as its mean user-perceived latency still remained 70 percent higher than that of Meta AI. More importantly, the same brevity pattern coincided with substantially lower quality on Open-Ended QA (28.0 percent vs. 72.0 percent score), where sufficient explanation and supporting details are essential. Effective glasses-oriented optimization must therefore jointly reduce pre-playback latency and adapt response length to task requirements, preserving concise answers for factual queries while allowing more elaborated responses for open-ended assistance.

Limitations and Future Work

Our study focuses on one RayBan device paired with Google Pixel 8a phones under controlled conditions. Extending the measurements to other glasses, companion phones, software versions, and network environments remains for future work. In addition, because the closed hardware does not allow the energy consumption of individual components to be isolated, controlled interface-level comparisons between BLE and Wi-Fi Direct also remain for future work. Our

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Live Streaming analysis presents a representative trace from ten repeated experiments to preserve the ordering of short-lived protocol events; future measurements will examine the frequency of cross-hop disruptions under broader network conditions. Finally, the QoE evaluation uses 100 English queries in a quiet environment and relies on two AI judges. Human evaluation and experiments involving noisy, multilingual, and multimodal interactions remain for future work.

Conclusion

This article presented a measurement-based case study of the networking architecture and AI QoE of RayBan. Our results show that current smartglasses employ function-specific interface choices, but repeated interface setup, independently maintained transport hops, and a fixed brevity-oriented AI strategy limit the resulting user experience. The broader lesson is that networking and AI should not be optimized in isolation. Future paired phones should evolve from connectivity relays into state-aware coordinators that anticipate workloads, reuse local links, and coordinate recovery across transport hops. Meanwhile, AI services should adapt response depth to task requirements while accounting for latency and wearable resource constraints. Such cross-layer, task-aware coordination can transform phone mediation from an additional system cost into an opportunity to jointly optimize connectivity, continuity, and AI QoE for future smartglasses.

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